1) a)

No vertical asymptotes as $x^2 + 3 \neq 0$

As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

So horizontal asymptote at $y = 1$

Asymptotes: $y = 1$

1) b)

No vertical asymptotes as $x^2 + 7 \neq 0$

As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

So horizontal asymptote at $y = 1$

Asymptotes: $y = 1$

1) c)

$x^2 = 2$ gives $x = \pm\sqrt{2}$

As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

So horizontal asymptote at $y = 1$

Asymptotes: $y = 1$, $x = -\sqrt{2}$ and $x = \sqrt{2}$

1) d)

$x^2 + 3x = 0$ gives $x = 0$ or $x = -3$

As $x \to \infty$, $y \to 3$ and as $x \to -\infty$, $y \to 3$

So horizontal asymptote at $y = 3$

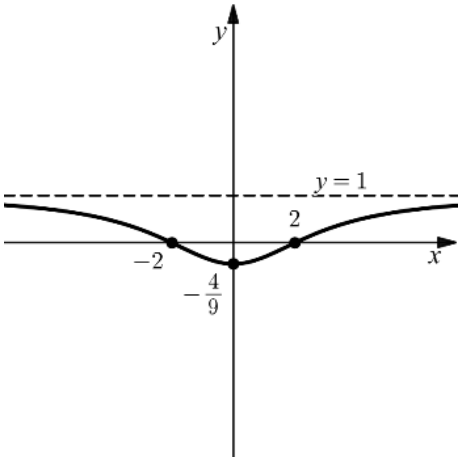
Asymptotes: $y = 3$, $x = 0$ and $x = -3$

2) a)

Asymptotes: $y = 1$

Intercepts: $\left(0, -\frac{4}{9}\right)$, $(2, 0)$, $(-2, 0)$

When x is a large positive, $y < 1$, when x is a large negative, $y < 1$

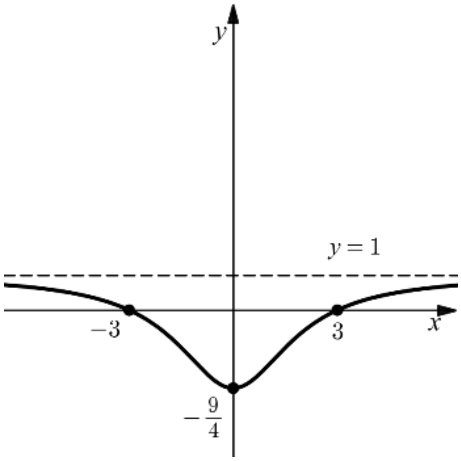


2) b)

Asymptotes: $y = 1$

Intercepts: $\left(0, -\frac{9}{4}\right)$, $(3, 0)$, $(-3, 0)$

When x is a large positive, $y < 1$, when x is a large negative, $y < 1$

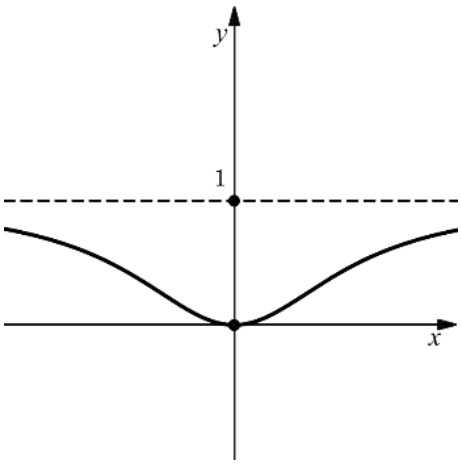


2) c)

Asymptotes: $y = 1$

Intercepts: $(0, 0)$

When x is a large positive, $y < 1$, when x is a large negative, $y < 1$



3) a)

$2(2x^2 + 3x + 9) = 2x^2 - 3x + 23$

$4x^2 + 6x + 18 = 2x^2 - 3x + 23$

$2x^2 + 9x - 5 = 0$

$(2x - 1)(x + 5) = 0$

$\therefore x = \frac{1}{2}$ or $x = -5$

3) b)

$x^2 - 2x - 5 = 2(x^2 - 4x + 2)$

$x^2 - 2x - 5 = 2x^2 - 8x + 4$

$x^2 - 6x + 9 = 0$

$(x - 3)^2 = 0$

$\therefore x = 3$

3) c)

$3(x^2 - 4x + 3) = 4(x^2 - 3x + 2)$

$3x^2 - 12x + 9 = 4x^2 - 12x + 8$

$x^2 - 1 = 0$

$(x - 1)(x + 1) = 0$

$x = 1$ or $x = -1$ but $x = 1$ gives 0 on the denominator

$\therefore x = -1$

Alternative Method:

$\frac{x^2-4x+3}{x^2-3x+2} = \frac{4}{3}$

$\Rightarrow \frac{(x-3)(x-1)}{(x-2)(x-1)} = \frac{4}{3}$

$\Rightarrow \frac{x-3}{x-2} = \frac{4}{3}$

$\Rightarrow 3x - 9 = 4x - 8$

$\therefore x = -1$

3) d)

$-(x^2 + 4) = 3(x^2 + 4x - 5)$

$-x^2 - 4 = 3x^2 + 12x - 15$

$4x^2 + 12x - 11 = 0$

$\therefore x = -\frac{3}{2} + \sqrt{5}$ or $x = -\frac{3}{2} - \sqrt{5}$

4) a)

Find the values for which y exists:

$k = \frac{x^2+2x+5}{x^2+2x-3} \Rightarrow k(x^2 + 2x - 3) = x^2 + 2x + 5$

$(k - 1)x^2 + (2k - 2)x - (3k + 5) = 0$

For k to exist: $(2k - 2)^2 - 4(k - 1)(-3k - 5) \geq 0$

$4k^2 - 8k + 4 + 12k^2 + 8k - 20 \geq 0$

$16k^2 - 16 \geq 0$

$(k - 1)(k + 1) \geq 0$ as req.

4) b)

$k = \pm 1$ from part a)

So $y > 1$ and $y \leq -1$

4) c)

P occurs when $y = -1$

5) a)

Find the values for which y exists:

$k = \frac{x^2-8x+4}{x^2-5x+4} \Rightarrow k(x^2 - 5x + 4) = x^2 - 8x + 4$

$(k-1)x^2 + (8-5k)x + (4k-4) = 0$

For k to exist: $(8-5k)^2 - 4(k-1)(4k-4) \geq 0$

$25k^2 - 80k + 64 - 16k^2 + 32k - 16 \geq 0$

$9k^2 - 48k + 48 \geq 0$

5) b)

$k = \frac{4}{3}$ or $k = 4$ from part a)

So $y \leq \frac{4}{3}$ and $y \geq 4$

5) c)

P occurs when $y = \frac{4}{3}$ and Q occurs when $y = 4$

6) a)

Find the values for which y exists:

$k = \frac{x^2+x+5}{x^2+2x-8} \Rightarrow k(x^2 + 2x - 8) = x^2 + x + 5$

$(k-1)x^2 + (2k-1)x - (8k+5) = 0$

For k to exist: $(2k-1)^2 - 4(k-1)(-8k-5) \geq 0$

$4k^2 - 4k + 1 + 32k^2 - 12k - 20 \geq 0$

$36k^2 - 16k - 19 \geq 0$ as req.

6) b)

$k = \frac{2}{9} \pm \frac{\sqrt{187}}{18}$ from part a)

Since $k = \frac{2}{9} + \frac{\sqrt{187}}{18} \approx 0.981 \ldots$ and $k = \frac{2}{9} - \frac{\sqrt{187}}{18} \approx -0.537 \ldots$

So $y > \frac{2}{9} + \frac{\sqrt{187}}{18}$ and $y \leq \frac{2}{9} - \frac{\sqrt{187}}{18}$

6) c)

P occurs when $y = \frac{2}{9} - \frac{\sqrt{187}}{18}$

7) a)

Find the values for which y exists:

$k = \frac{x^2-2x}{x^2-8x+16} \Rightarrow k(x^2 - 8x + 16) = x^2 - 2x$

$(k-1)x^2 + (2-8k)x + 16k = 0$

For k to exist: $(2-8k)^2 - 4(k-1)16k \geq 0$

$64k^2 - 32k + 4 - 64k^2 + 64k \geq 0$

$32k + 4 \geq 0$ as req.

7) b)

$k = -\frac{1}{8}$ from part a)

$y \geq -\frac{1}{8}$

7) c)

Q occurs when $y = -\frac{1}{8}$

8) a)

$x^2 + 5 > 0$ for all real values, so is not equal to zero

8) b)

$y = 0$

8) c)

$k = \frac{x+2}{x^2+5} \Rightarrow k(x^2 + 5) = x + 2$

$kx^2 + 5k = x + 2$

$kx^2 - x + 5k - 2 = 0$

$kx^2 - x + (5k - 2) = 0$ as req.

8) d)

For k to exist: $(-1)^2 - 4k(5k-2) \geq 0$

$-20k^2 + 8k + 1 \geq 0$

$k = -\frac{1}{10}$ or $k = \frac{1}{2}$

So $-\frac{1}{10} \leq k \leq \frac{1}{2}$

8) e)

$-\frac{1}{10} = \frac{x+2}{x^2+5} \Rightarrow -\frac{1}{10}x^2 - x + \left(5 \times -\frac{1}{10} - 2\right) = 0$

$x^2 + 10x + 5 = (x+5)^2 = 0 \therefore x = -5$

$\frac{1}{2} = \frac{x+2}{x^2+5} \Rightarrow \frac{1}{2}x^2 - x + \left(5 \times \frac{1}{2} - 2\right) = 0$

$x^2 - 2x + 1 = (x-1)^2 = 0 \therefore x = 1$

So $\left(-5, -\frac{1}{10}\right)$ and $\left(1, \frac{1}{2}\right)$

9) a)

Find the values for which k exists:

$k = \frac{12x}{x^2+4} \Rightarrow k(x^2+4) = 12x$

$kx^2 - 12x + 4k = 0$

For k to exist: $(-12)^2 - 16k^2 \geq 0$

$144 - 16k^2 \geq 0$

$(k+3)(k-3) \geq 0$

So $-3 \leq k \leq 3$

And so $-3 \leq \frac{12x}{x^2+4} \leq 3$

9) b)

Intercept: $(0, 0)$

Asymptote: $y = 0$

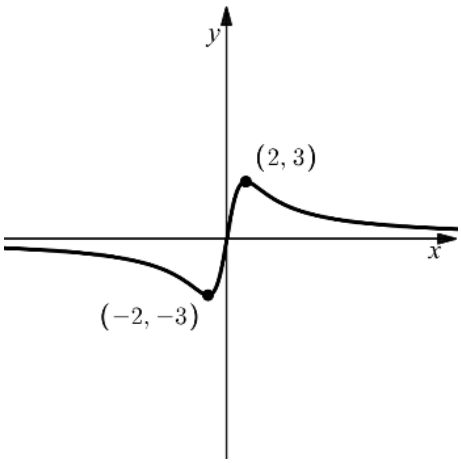
The graph exists for $-3 \leq y \leq 3$

When $y = 3$, $3 = \frac{12x}{x^2+4} \Rightarrow 3(x^2 + 4) = 12x \Rightarrow 3x^2 - 12x + 12 = 0$ so $x = 2$

Max is at $(2, 3)$

When $y = -3$, $-3 = \frac{12x}{x^2+4} \Rightarrow -3(x^2 + 4) = 12x \Rightarrow 3x^2 + 12x + 12 = 0$ so $x = -2$

Min is at $(-2, -3)$



9) c)

g(x) is the reciprocal of f(x), so g(x) can have the reciprocal values of f(x), that is, between $-\frac{1}{3}$ and $\frac{1}{3}$

However, g(x) is undefined for $x = 0$

10) a)

Asymptotes: $y = 1$, $x = -1$, $x = 1$

Intercepts: $(0, -4)$

When x is a large positive, $y > 1$, when x is a large negative, $y > 1$

Find the values for which y exists:

$k = \frac{x^2+4}{x^2-1} \Rightarrow k(x^2 - 1) = x^2 + 4$

$(k - 1)x^2 + (-k - 4) = 0$

For k to exist: $0^2 - 4(k - 1)(-k - 4) \geq 0$

$4k^2 + 12k - 16 \geq 0$

$k = -4$ or $k = 1$

So $k \leq -4$ and $k > 1$

When $y = -4$, $-4 = \frac{x^2+4}{x^2-1} \Rightarrow -4(x^2 - 1) = x^2 + 4 \Rightarrow 0 = 5x^2$ so $x = 0$

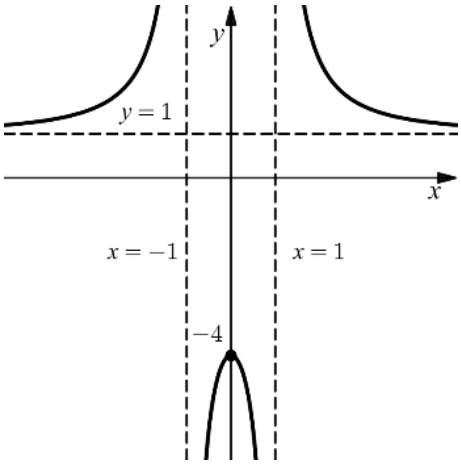
Local max is at $(0, -4)$

At $x = -1^-$, y is large positive

At $x = -1^+$, y is large negative

At $x = 1^-$, y is large negative

At $x = 1^+$, y is large positive



10) b)

Asymptotes: $y = 1$, $x = -4$, $x = 0$

$x^2 - 4x + 4 = 0 \Rightarrow x = 2$ so intercepts: $(2, 0)$

When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

Find the values for which y exists:

$k = \frac{x^2-4x+4}{x^2+4x} \Rightarrow k(x^2 + 4x) = x^2 - 4x + 4$

$(k - 1)x^2 + (4k + 4)x - 4 = 0$

For k to exist: $(4k + 4)^2 - 4(k - 1)(-4) \geq 0$

$16k^2 + 32k + 16 + 16k - 16 \geq 0$

$16k^2 + 48k \geq 0$

$k = -3$ or $k = 0$

So $k \leq -3$ and $k \geq 0$

When $y = -3$, $-3 = \frac{x^2-4x+4}{x^2+4x} \Rightarrow -3(x^2 + 4x) = x^2 - 4x + 4$

$\Rightarrow -3x^2 - 12x = x^2 - 4x + 4$

$\Rightarrow 0 = 4x^2 + 8x + 4 = 4(x + 1)^2$ so $x = -1$

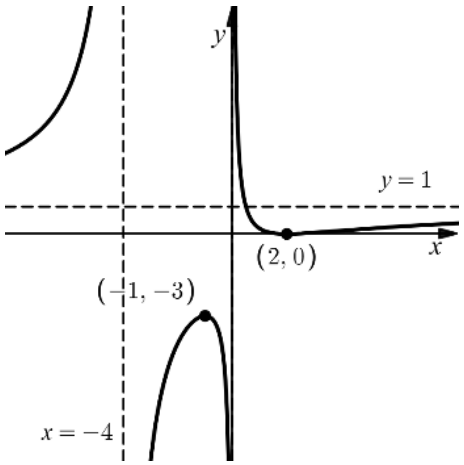
Local max is at $(-1, -3)$

At $x = -1^-$, y is large positive

At $x = -1^+$, y is large negative

At $x = 0^-$, y is large negative

At $x = 0^+$, y is large positive



10) c)

$x^2 + 2x + 2 = 0$ has no real solutions, so no vertical asymptotes.

Asymptotes: $y = 1$

$x^2 + 8x + 16 = 0 \Rightarrow x = -4$ so intercepts: $(-4, 0)$ and $(0, 8)$

When $x \to \infty$, $y > 1$, when $x \to -\infty$, $y < 1$

Find the values for which y exists:

$k = \frac{x^2+8x+16}{x^2+2x+2} \Rightarrow k(x^2 + 2x + 2) = x^2 + 8x + 16$

$(k-1)x^2 + (2k-8)x + (2k-16) = 0$

For k to exist: $(2k-8)^2 - 4(k-1)(2k-16) \geq 0$

$4k^2 - 32k + 64 - 8k^2 + 72k - 64 \geq 0$

$-4k^2 + 40k \geq 0$

$k = 0$ or $k = 10$

So $k \leq 10$ and $k \geq 0$

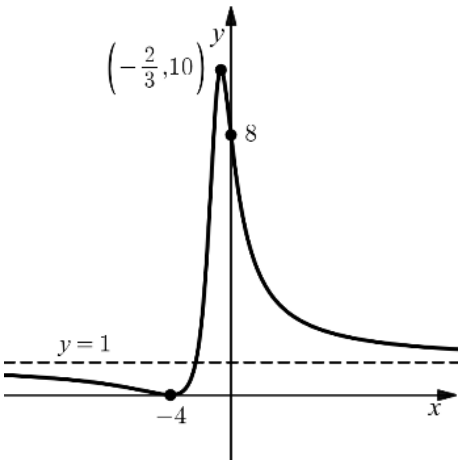
When $y = 10$, $10 = \frac{x^2+8x+16}{x^2+2x+2} \Rightarrow 10(x^2 + 2x + 2) = x^2 + 8x + 16$

$\Rightarrow 9x^2 + 12x + 4 = 0$ so $x = -\frac{2}{3}$

Local max is at $\left(-\frac{2}{3}, 10\right)$

When $y = 0$, $0 = \frac{x^2+8x+16}{x^2+2x+2} \Rightarrow x = -4$

Local min is at $(-4, 0)$



10) d)

$x^2 + 2 = 0$ has no real solutions, so no vertical asymptotes.

Asymptotes: $y = 1$

$x^2 + 4x = 0 \Rightarrow x = -4$ so intercepts: $(-4, 0)$ and $(0, 0)$

When $x \to \infty$, $y > 1$, when $x \to -\infty$, $y < 1$

Find the values for which y exists:

$k = \frac{x^2+4x}{x^2+2} \Rightarrow k(x^2 + 2) = x^2 + 4x$

$(k - 1)x^2 - 4x + 2k = 0$

For k to exist: $(-4)^2 - 4(k - 1)2k \geq 0$

$16 + 8k - 8k^2 \geq 0$

$k = -1$ or $k = 2$

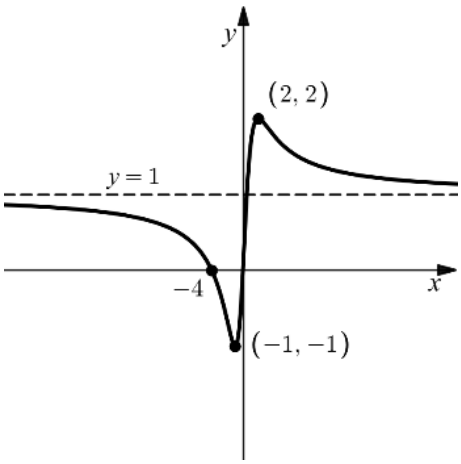
So $k \leq 2$ and $k \geq -1$

When $y = 2$, $2 = \frac{x^2+4x}{x^2+2} \Rightarrow 2(x^2 + 2) = x^2 + 4x \Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$

Local max is at $(2, 2)$

When $y = -1$, $-1 = \frac{x^2+4x}{x^2+2} \Rightarrow -(x^2 + 2) = x^2 + 4x \Rightarrow 2x^2 + 4x + 2 = 0$ so $x = -1$

Local min is at $(-1, 1)$



10) e)

Asymptotes: $y = 1$, $x = 0$, $x = -4$

$x^2 - 4x - 5 = 0 \Rightarrow x = 5$, $x = -1$ so intercepts: $(5, 0)$ and $(-1, 0)$

No vertical intercept.

When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

Find the values for which y exists:

$k = \frac{x^2-4x-5}{x^2+4x} \Rightarrow k(x^2 + 4x) = x^2 - 4x - 5$

$(k - 1)x^2 + (4k + 4)x + 5 = 0$

For k to exist: $(4k + 4)^2 - 4(k - 1)5 \geq 0$

$16k^2 + 32k + 16 - 20k + 20 \geq 0$

$16k^2 + 12k + 36 \geq 0$

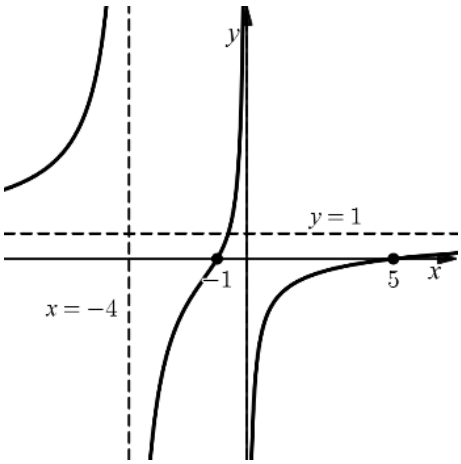
Which is true for all values of k, so y exists for all values (except $y = 1$).

At $x = -4^-$, y is large positive

At $x = -4^+$, y is large negative

At $x = 0^-$, y is large positive

At $x = 0^+$, y is large negative



10) f)

$x^2 - 4x + 3 = 0 \Rightarrow x = 1, x = 3$

Asymptotes: $y = 1$, $x = 1$, $x = 3$

$x^2 - 4x + 2 = 0 \Rightarrow x = 2 - \sqrt{2}, x = 2 + \sqrt{2}$ so intercepts: $\left(2 - \sqrt{2}, 0\right)$ and $\left(2 + \sqrt{2}, 0\right)$

Vertical intercept at $\left(0, \frac{2}{3}\right)$

When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

Find the values for which y exists:

$k = \frac{x^2-4x+2}{x^2-4x+3} \Rightarrow k(x^2 - 4x + 3) = x^2 - 4x + 2$

$(k - 1)x^2 + (4 - 4k)x + (3k - 2) = 0$

For k to exist: $(4 - 4k)^2 - 4(k - 1)(3k - 2) \geq 0$

$16k^2 - 32k + 16 - 12k^2 + 20k - 8 \geq 0$

$4k^2 - 12k + 8 \geq 0$

$k = 1$ or $k = 2$

So $k < 1$ and $k \geq 2$

When $y = 2$, $2 = \frac{x^2-4x+2}{x^2-4x+3} \Rightarrow 2(x^2 - 4x + 3) = x^2 - 4x + 2$

$\Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$

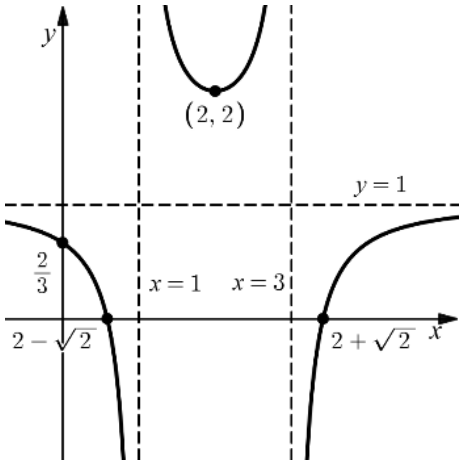
Local min is at $(2, 2)$

At $x = 1^-$, y is large negative

At $x = 1^+$, y is large positive

At $x = 3^-$, y is large positive

At $x = 3^+$, y is large negative



11) a)

Horizontal asymptote: $y = 2$

Vertical asymptote: none as $x^2 + 2x + 5 > 0$

$\therefore y = 2$

11) b)

Horizontal asymptote: $y = 3$

Vertical asymptote: none as $x^2 + 2x + 3 > 0$

$\therefore y = 3$

11) c)

Horizontal asymptote: $y = \frac{1}{4}$

Vertical asymptote: $4x^2 - 8x + 4 = 0$ so $x = 1$

$\therefore y = \frac{1}{4}$, $x = 1$

11) d)

Horizontal asymptote: $y = \frac{5}{4}$

Vertical asymptote: $2x^2 + 7x - 15 = 0$ so $x = -5$ or $x = \frac{3}{2}$

$\therefore y = \frac{5}{4}$, $x = -5$, $x = \frac{3}{2}$

12)

$y = a$, $x = \sqrt{d}$, $x = -\sqrt{d}$

13) a)

$y = 4$

13) b)

Find the values for which y exists:

$k = \frac{4x^2+4x+11}{x^2-x+1} \Rightarrow k(x^2 - x + 1) = 4x^2 + 4x + 11$

$(k-4)x^2 + (-k-4)x + (k-11) = 0$

For k to exist: $(-k-4)^2 - 4(k-4)(k-11) \geq 0$

$k^2 + 8k + 16 - 4k^2 + 60k - 176 \geq 0$

$-3k^2 + 68k - 160 \geq 0$

$3k^2 - 68k + 160 \leq 0$

$(3k-8)(k-20) \leq 0$ as req.

13) c)

The graph exists for $\frac{8}{3} \leq y \leq 20$

When $y = 20$, $20 = \frac{4x^2+4x+11}{x^2-x+1} \Rightarrow 20(x^2 - x + 1) = 4x^2 + 4x + 11$

$\Rightarrow 16x^2 - 24x + 9 = 0$ so $x = \frac{3}{4}$

Max is at $\left(\frac{3}{4}, 20\right)$

When $y = \frac{8}{3}$, $\frac{8}{3} = \frac{4x^2+4x+11}{x^2-x+1} \Rightarrow \frac{8}{3}(x^2 - x + 1) = 4x^2 + 4x + 11$

$\Rightarrow 8x^2 - 8x + 8 = 12x^2 + 12x + 33$

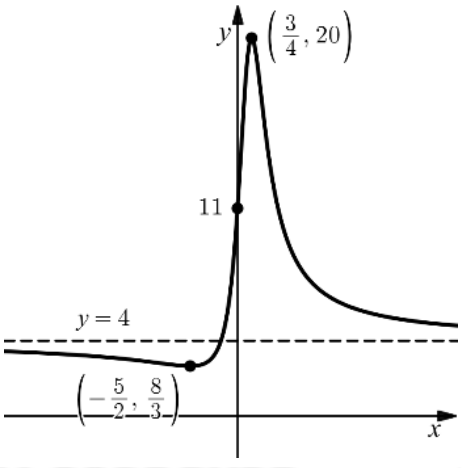
$\Rightarrow 4x^2 + 20x + 25 = 0$ so $x = -\frac{5}{2}$

Min is at $\left(-\frac{5}{2}, \frac{8}{3}\right)$

$\therefore$ Max at $\left(\frac{3}{4}, 20\right)$ and min at $\left(-\frac{5}{2}, \frac{8}{3}\right)$ as the graph exists for $\frac{8}{3} \leq y \leq 20$

13) d)

Intercept at $(0, 11)$



14) a)

$p = 2$

14) b)

$x = 2, x = -2, y = 2$

14) c)

Find the values for which y exists:

$k = \frac{2x^2-4x+9}{x^2-4} \Rightarrow k(x^2-4) = 2x^2 - 4x + 9$

$(k-2)x^2 + 4x + (-4k-9) = 0$

For k to exist: $4^2 - 4(k-2)(-4k-9) \geq 0$

$16 + 16k^2 + 4k - 72 \geq 0$

$4k^2 + k - 14 \geq 0$

$(4k-7)(k+2) \geq 0$ as req.

14) d)

The graph exists for $y \leq -2$ and $y \geq \frac{7}{4}$ (which is below the asymptote)

When $y = -2$, $-2 = \frac{2x^2-4x+9}{x^2-4} \Rightarrow -2(x^2-4) = 2x^2 - 4x + 9$

$\Rightarrow 4x^2 - 4x + 1 = 0$ so $x = \frac{1}{2}$

Local max is at $\left(\frac{1}{2}, -2\right)$

When $y = \frac{7}{4}$, $\frac{7}{4} = \frac{2x^2-4x+9}{x^2-4} \Rightarrow \frac{7}{4}(x^2-4) = 2x^2 - 4x + 9$

$\Rightarrow 7x^2 - 28 = 8x^2 - 16x + 36$

$\Rightarrow x^2 - 16x + 64 = 0$ so $x = 8$

Local min is at $\left(8, \frac{7}{4}\right)$

∴ Local max is at $\left(\frac{1}{2}, -2\right)$ and local min is at $\left(8, \frac{7}{4}\right)$

14) e)

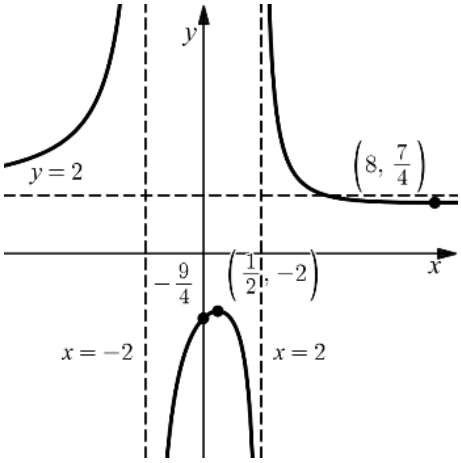
Intercept at $\left(0, -\frac{9}{4}\right)$

At $x = -2^-$, y is large positive

At $x = -2^+$, y is large negative

At $x = 2^-$, y is large negative

At $x = 2^+$, y is large positive



15) a)

$y = \frac{1}{3}$

15) b)

Find the values for which y exists:

$k = \frac{x^2-8x+1}{3x^2+1} \Rightarrow k(3x^2+1) = x^2 - 8x + 1$

$(3k-1)x^2 + 8x + (k-1) = 0$

For k to exist: $8^2 - 4(3k-1)(k-1) \geq 0$

$64 - 12k^2 + 16k - 4 \geq 0$

$-12k^2 + 16k + 60 \geq 0$

$3k^2 - 4k - 15 \leq 0$

$(3k+5)(k-3) \leq 0$ as req.

15) c)

The graph exists for $-\frac{5}{3} \leq y \leq 3$

When $y = 3$, $3 = \frac{x^2-8x+1}{3x^2+1} \Rightarrow 3(3x^2+1) = x^2 - 8x + 1$

$\Rightarrow 8x^2 + 8x + 2 = 0$ so $x = -\frac{1}{2}$

Max is at $\left(-\frac{1}{2}, 3\right)$

When $y = -\frac{5}{3}$, $-\frac{5}{3} = \frac{x^2-8x+1}{3x^2+1} \Rightarrow -\frac{5}{3}(3x^2+1) = x^2 - 8x + 1$

$\Rightarrow -15x^2 - 5 = 3x^2 - 24x + 3$

$\Rightarrow 18x^2 - 24x + 8 = 0$ so $x = \frac{2}{3}$

Min is at $\left(\frac{2}{3}, -\frac{5}{3}\right)$

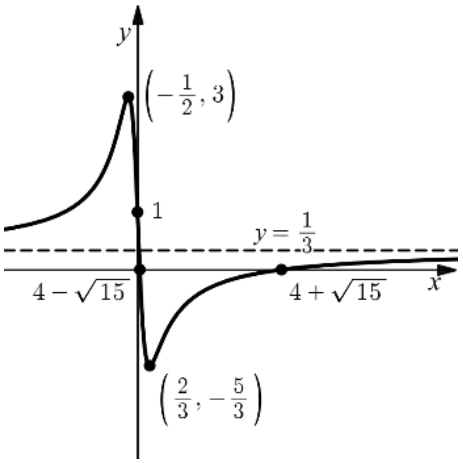
$\therefore$ Max is at $\left(-\frac{1}{2}, 3\right)$ and min is at $\left(\frac{2}{3}, -\frac{5}{3}\right)$

15) d)

Intercept at $(0,1)$

$x^2 - 8x + 1 = 0$ so $x = 4 \pm \sqrt{15}$

But since the curve tends to $y = \frac{1}{3}$ it only crosses the curve once and it is for a larger x-value than $x = \frac{2}{3}$



16) a)

Asymptote: $y = 0$

Find the values for which y exists:

$k = \frac{1}{2x^2+3} \Rightarrow k(2x^2 + 3) = 1$

$2kx^2 + (3k - 1) = 0$

For k to exist: $0^2 - 4(2k)(3k - 1) \geq 0$

$-24k^2 + 8k \geq 0$

$8k(3k - 1) \leq 0$

The graph exists for $0 < y \leq \frac{1}{3}$

When $y = \frac{1}{3}$, $\frac{1}{3} = \frac{1}{2x^2+3} \Rightarrow \frac{1}{3}(2x^2 + 3) = 1$

$\Rightarrow 2x^2 = 0$ so $x = 0$

Max is at $\left(0, \frac{1}{3}\right)$

∴ Max is at $\left(0, \frac{1}{3}\right)$ and asymptote is $y = 0$

16) b)

Asymptote: $y = 0$

Find the values for which y exists:

$k = \frac{1}{3(x^2+2x+2)} \Rightarrow 3k(x^2 + 2x + 2) = 1$

$3kx^2 + 6kx + (6k - 1) = 0$

For k to exist: $(6k)^2 - 4(3k)(6k - 1) \geq 0$

$36k^2 - 72k^2 + 12k \geq 0$

$3k^2 - k \leq 0$

$k(3k - 1) \leq 0$

The graph exists for $0 < y \leq \frac{1}{3}$

When $y = \frac{1}{3}$, $\frac{1}{3} = \frac{1}{3(x^2+2x+2)} \Rightarrow x^2 + 2x + 2 = 1$

$\Rightarrow x^2 + 2x + 1 = 0$ so $x = -1$

Max is at $\left(-1, \frac{1}{3}\right)$

∴ Max is at $\left(-1, \frac{1}{3}\right)$ and asymptote is $y = 0$

17)

Since $g(x)$ is the reciprocal of $f(x)$ it will become a minimum point as long as $b \neq 0$

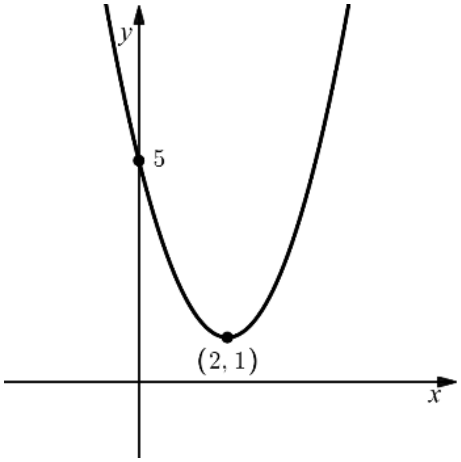
18) a)

$x^2 - 4x + 5 = (x-2)^2 + 1$

So minimum point is at $(2, 1)$

18) b)

Intercept: $(0, 5)$



18) c)

Reciprocal of previous minimum:

When $y = \frac{1}{1} = 1$, $1 = \frac{1}{x^2-4x+5} \Rightarrow x^2 - 4x + 5 = 1$

$\Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$

$\therefore$ Max is at $(2, 1)$

Alternative Method:

Find the values for which y exists:

$k = \frac{1}{x^2-4x+5} \Rightarrow k(x^2 - 4x + 5) = 1$

$kx^2 - 4kx + (5k - 1) = 0$

For k to exist: $(-4k)^2 - 4k(5k - 1) \geq 0$

$16k^2 - 20k^2 + 4k \geq 0$

$k^2 - k \leq 0$

$k(k - 1) \leq 0$

The graph exists for $0 < y \leq 1$ (since asymptote at $y = 0$)

When $y = 1$, $1 = \frac{1}{x^2-4x+5} \Rightarrow x^2 - 4x + 5 = 1$

$\Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$ $\therefore$ Max is at $(2, 1)$

18) d)

Since both graphs have the same stationary point, the tangent line is $y = 1$

19) a)

$k = \frac{3x^2-8x+7}{x^2+4x-6} \Rightarrow k(x^2 + 4x - 6) = 3x^2 - 8x + 7$

$(k - 3)x^2 + (4k + 8)x + (-6k - 7) = 0$

For k to exist: $(4k + 8)^2 - 4(k - 3)(-6k - 7) \geq 0$

$16k^2 + 64k + 64 + 24k^2 - 44k - 84 \geq 0$

$40k^2 + 20k - 20 \geq 0$

$(2k - 1)(k + 1) \geq 0$ as req.

19) b)

$k \leq -1$ or $k \geq \frac{1}{2}$

The graph exists for $y \leq -1$ and $y \geq \frac{1}{2}$

When $y = -1$, $-1 = \frac{3x^2-8x+7}{x^2+4x-6} \Rightarrow -(x^2+4x-6) = 3x^2-8x+7$

$\Rightarrow 4x^2-4x+1 = 0$ so $x = \frac{1}{2}$

Local max is at $\left(\frac{1}{2}, -1\right)$

When $y = \frac{1}{2}$, $\frac{1}{2} = \frac{3x^2-8x+7}{x^2+4x-6} \Rightarrow \frac{1}{2}(x^2+4x-6) = 3x^2-8x+7$

$\Rightarrow x^2+4x-6 = 6x^2-16x+14$

$\Rightarrow 5x^2-20x+20 = 0$ so $x = 2$

Local Min is at $\left(2, \frac{1}{2}\right)$

∴ Local max is at $\left(\frac{1}{2}, -1\right)$ and local min is at $\left(2, \frac{1}{2}\right)$

19) c)

$y = \frac{x^2+4x-6}{3x^2-8x+7}$

$3x^2-8x+7 = 0$ has no real solutions (the solutions are $\frac{4}{3} \pm \frac{\sqrt{5}}{3}$i)

∴ no vertical asymptotes.

19) d)

Since $\mathrm{f}(x) \geq \frac{1}{2}$ and $\mathrm{f}(x) \leq -1$ then $-1 \leq \frac{1}{\mathrm{f}(x)} \leq 2$

So minimum of $\frac{1}{\mathrm{f}(x)}$ is when $\frac{1}{\mathrm{f}(x)} = -1$

$-1 = \frac{x^2+4x-6}{3x^2-8x+7} \Rightarrow -(3x^2-8x+7) = x^2+4x-6$

$\Rightarrow 4x^2-4x+1 = 0$ so $x = \frac{1}{2}$

So it has a minimum at $\left(\frac{1}{2}, -1\right)$ which is the local maximum of $\mathrm{f}(x)$

So the tangent line must be $y = -1$

20) a)

$y = -1$

20) b)

No solutions to $x^2 + kx + 2 = 0$

$k^2 - 4 \times 2 < 0$

$\left(k - 2\sqrt{2}\right)\left(k + 2\sqrt{2}\right) < 0$

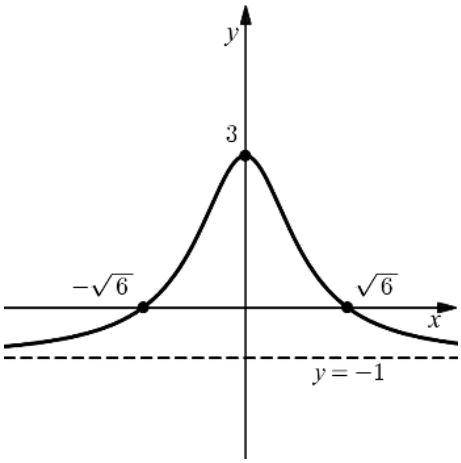
So $-2\sqrt{2} < k < 2\sqrt{2}$

20) c)

$(0, 3)$, $\left(\sqrt{6}, 0\right)$ and $\left(-\sqrt{6}, 0\right)$

20) d)

The curve is symmetrical about the y-axis when $k = 0$



21) a)

$a = 4$

21) b)

$x^2 + bx - 8 = 0$

$(x - 4)(x + 2) = 0$

$x^2 - 2x - 8 = 0$

So $b = -2$

21) c)

$x = -2$

21) d)

$y = \frac{4x^2-8x+8}{x^2-2x-8}$

$(0, -1)$

21) e)

Find the values for which y exists:

$k = \frac{4x^2-8x+8}{x^2-2x-8} \Rightarrow k(x^2 - 2x - 8) = 4x^2 - 8x + 8$

$(k-4)x^2 + (8-2k)x + (-8k-8) = 0$

For k to exist: $(8-2k)^2 - 4(k-4)(-8k-8) \geq 0$

$4k^2 - 32k + 64 + 32k^2 - 96k - 128 \geq 0$

$36k^2 - 128k - 64 \geq 0$

$(9k-4)(k-4) \geq 0$

The graph exists for $y \leq -\frac{4}{9}$ or $y > 4$

When $y = -\frac{4}{9}$, $-\frac{4}{9} = \frac{4x^2-8x+8}{x^2-2x-8} \Rightarrow -4(x^2 - 2x - 8) = 9(4x^2 - 8x + 8)$

$\Rightarrow x^2 - 2x + 1 = 0$ so $x = 1$

$\therefore$ Local max is at $\left(1, -\frac{4}{9}\right)$

22) a)

$a = 2$

When $y = 0$, $x = -1 \pm \sqrt{\frac{3}{2}}$

$2x^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 8c}}{4}$

So $b = 4$

$\frac{\sqrt{16-8c}}{4} = \sqrt{\frac{3}{2}} \Rightarrow \frac{16-8c}{16} = \frac{3}{2} \Rightarrow 16 - 8c = 24 \therefore c = -1$

When $x = 0$, $y = -1$ so $\frac{-1}{q} = -1 \therefore q = 1$

$y = \frac{2x^2+4x-1}{x^2+px-1}$

At $\left(-\frac{1}{2}, -2\right)$

$-2 = \frac{2\left(-\frac{1}{2}\right)^2 + 4\left(-\frac{1}{2}\right) - 1}{\left(-\frac{1}{2}\right)^2 + p\left(-\frac{1}{2}\right) - 1} \Rightarrow -2 = \frac{-\frac{5}{2}}{-\frac{1}{2}p - \frac{5}{4}} \Rightarrow p + \frac{10}{4} = -\frac{5}{2} \therefore p = 0$

$\mathrm{f}(x) = \frac{2x^2+4x-1}{x^2+1}$

22) b)

Find the values for which y exists:

$k = \frac{2x^2+4x-1}{x^2+1} \Rightarrow k(x^2 + 1) = 2x^2 + 4x - 1$

$(k-2)x^2 - 4x + (k+1) = 0$

For k to exist: $(-4)^2 - 4(k-2)(k+1) \geq 0$

$16 - 4k^2 + 4k + 8 \geq 0$

$k^2 - k - 6 \leq 0$

$(k+2)(k-3) \leq 0$

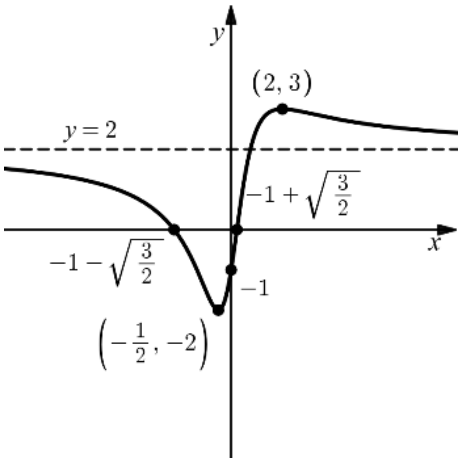
The graph exists for $-2 \leq y \leq 3$

When $y = 3$, $3 = \frac{2x^2+4x-1}{x^2+1} \Rightarrow 3(x^2 + 1) = 2x^2 + 4x - 1$

$\Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$

$\therefore$ Local max is at $(2, 3)$

22) c)



23) a)

$a = p$

23) b)

When $x = 0$, $y = 2$ so $\frac{c}{r} = 2$

$\therefore c = 2r$

23) c)

Since $a = 2$

When $y = 0$, $x = 2$ or $x = -\frac{3}{2}$

So the numerator is $k(x - 2)(2x + 3)$ is for some value of k

$k(2x^2 - x - 6)$

So $k = 1$

$\mathrm{f}(x) = \frac{2x^2 - x - 6}{2x^2 + qx + r}$

But since $c = 2r$ then $r = -3$

$\mathrm{f}(x) = \frac{2x^2 - x - 6}{2x^2 + qx - 3}$

When $x = -3$, $2x^2 + qx - 3 = 0$ so $18 - 3q - 3 = 0 \therefore q = 5$

$\mathrm{f}(x) = \frac{2x^2 - x - 6}{2x^2 + 5x - 3}$

23) d)

$2x^2 + 5x - 3 = 0$ has solutions $x = -3$ and $x = \frac{1}{2}$

So the other asymptote is $x = \frac{1}{2}$

24)

Find the values for which y exists:

$k = \frac{x^2+p}{x^2-q^2} \Rightarrow k(x^2 - q^2) = x^2 + p$

$(k-1)x^2 + (-kq^2 - p) = 0$

For k to exist: $0^2 - 4(k-1)(-kq^2 - p) \geq 0$

$\Rightarrow (k-1)(-kq^2 - p) \leq 0$

$\Rightarrow (k-1)(q^2k + p) \geq 0$

$\Rightarrow k \leq -\frac{p}{q^2}$ or $k \geq 1$

The graph exists for $y \leq -\frac{p}{q^2}$ and $y \geq 1$

25) a)

No vertical asymptotes.

$y = 1$

25) b)

Find the values for which y exists:

$k = \frac{x^2-px+p}{x^2+q} \Rightarrow k(x^2+q) = x^2 - px + p$

$(k-1)x^2 + px + (kq-p) = 0$

For k to exist: $p^2 - 4(k-1)(kq-p) \geq 0$

$p^2 - 4qk^2 + 4(p+q)k - 4p \geq 0$

$4qk^2 - 4(p+q)k + (4p-p^2) \leq 0$

Solving $4qk^2 - 4(p+q)k + (4p-p^2) = 0$ gives

$k = \frac{4(p+q) \pm \sqrt{(-4(p+q))^2 - 4(4q)(4p-p^2)}}{8q}$

$= \frac{4(p+q) \pm \sqrt{16((p+q))^2 - 16q(4p-p^2)}}{8q}$

$= \frac{4(p+q) \pm 4\sqrt{(p+q)^2 - q(4p-p^2)}}{8q}$

$= \frac{p+q \pm \sqrt{p^2+2pq+q^2-4pq+p^2q}}{2q}$

$= \frac{p+q \pm \sqrt{p^2-2pq+q^2+p^2q}}{2q}$

So the graph does not exist for $y < \frac{p+q-\sqrt{p^2-2pq+q^2+p^2q}}{2q}$ and $y > \frac{p+q \pm \sqrt{p^2-2pq+q^2+p^2q}}{2q}$